Functional Requirements (HVPN-FR-NNN)

Functional Requirements (HVPN-FR-NNN)

Revision: 2 **Last modified:** 2026-06-26T12:00:00Z

> **Reconciled (§11.4.35, 2026-06-26):** FR-101's RBAC role set is aligned to the authoritative `svc-identity.md` enum `{member, operator, admin}` (no "tenant owner"). FR-019 is restated as a fixed asymmetric per-leg default for MVP with map-driven per-leg policy deferred to P2. Owning-doc citations to the non-existent "Connector spec" / "edge spec" are repointed to concrete files (`v04-client/helix-core-rust.md` advertise/route mode + `v03-control-plane/svc-registry.md` for the connector; `01-data-plane.md` for the edge).

> **Document role.** Volume 1 deep document that enumerates **every functional requirement** of HelixVPN as a numbered, traceable list (`HVPN-FR-NNN`), grouped by capability area. It is the requirements spine the components and tests trace back to (the planned `../v00-meta/requirements-traceability.md` closes the FR → component → test loop). It deepens the capability set of the overview `00-product-scope-and-principles.md` — the Mullvad-parity matrix (§9), the differentiators (X1–X5), the seven principles (§8), and the 8-criteria MVP DoD (§10.2) — into testable statements.
>
> Non-functional requirements (perf/scale/availability/privacy/security SLOs) are a sibling document (`nonfunctional-requirements.md`, planned, `HVPN-NFR-NNN`); this document owns *functional* behaviour only. Where a requirement carries a quantitative target that is really an SLO (e.g. p99 < 1 s convergence), it is stated here as the functional acceptance gate and cross-referenced to the NFR doc.
>
> **Status:** SPEC-ONLY. Each FR's *satisfying component* names the Volume 2–6 doc that implements it; if that doc has not yet fixed the detail, the FR is the binding intent and the detail is `UNVERIFIED` (§11.4.6). Quantitative targets are quoted from the overview/spine/security-overview, never invented.

> **Convention.** RFC 2119 force (MUST/SHOULD/MAY) per [SPEC §0]. Priority ∈ {**MVP** (Phase 1), **P2** (Phase 2), **P3** (Phase 3)} per the phase scope [00 §10], [SPEC §8]. Each FR row: **ID · Statement · Acceptance criterion · Satisfying component (owning doc) · Priority**. Acceptance criteria requiring captured runtime evidence (§11.4.69) are flagged `[evidence]`.

Table of contents

---

0. Numbering, priority & traceability

- IDs are stable and never reused (`HVPN-FR-NNN`, zero-padded). Areas are allocated number bands (A=0xx, B=1xx, … L=11xx) so insertions stay grouped.
- Every FR has exactly one **owning doc** (the Volume 2–6 spec that implements + tests it). The FR is the *what*; the owning doc is the *how*.
- Priority follows the phase boundary [00 §10]; an FR marked **P2/P3** MUST NOT be built before its phase (it is a tracked §11.4.93 workable item until then).
- Acceptance criteria are deliberately *falsifiable*: a metadata-only / config-only PASS is forbidden (§11.4 / §11.4.69); `[evidence]` marks criteria that require captured runtime evidence.

```
flowchart LR
  FR["HVPN-FR-NNN<br/>(this doc)"] --> COMP["Satisfying component<br/>(Volume 2–6 doc)"]
  COMP --> TEST["Test type<br/>(Volume 8, §11.4.169)"]
  TEST --> EV["Captured evidence<br/>(§11.4.69)"]
  FR -.priority.-> PHASE["Phase boundary<br/>(00 §10 / SPEC §8)"]
  FR -.traces to.-> DOD["8-criteria MVP DoD<br/>(§M)"]
```

A. Connect & transport (FR-0xx)

Owning docs: `../01-data-plane.md` and Volume 2 (`transport-*.md`, `transport-selection-ladder.md`, `obfuscation-and-dpi.md`). Parity rows F1–F9 [00 §9].

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-001	The data channel MUST use WireGuard (Noise IK, Curve25519, ChaCha20-Poly1305) as the cryptographic core; obfuscation is a layer **beneath** WG, never a crypto fork.	A CI lint forbids edits to WG crypto primitives; a tunnel establishes with stock WG crypto. `[evidence]` handshake capture.	`01`, `v02-data-plane/wireguard-core.md`	MVP
HVPN-FR-002	A Client MUST establish a plain-UDP WireGuard tunnel to the Gateway (fast path) reaching ≥80% of bare-link throughput.	G1: iperf3 capture client→gw→connector LAN ≥80% bare link. `[evidence]`	`v02-data-plane/transport-plain-udp.md`	MVP
HVPN-FR-003	The data plane MUST expose obfuscation modes through a single `Transport` trait (one implementation, three consumers: client, connector, edge).	`Transport` trait compiles; client+edge share the same crate byte-for-byte.	`01` (§7.1 trait), `v02-data-plane/transport-trait.md`	MVP
HVPN-FR-004	The Client MUST support MASQUE/QUIC (WG-over-HTTP-3, RFC 9298/9297/9221) obfuscation that survives a DPI UDP block.	G2: MASQUE survives an nftables UDP block at ≥50% of plain-WG throughput. `[evidence]`	`v02-data-plane/transport-masque-quic.md`	MVP
HVPN-FR-005	The Client MUST support lightweight obfuscation (LWO — keyed WG-header obfs + padding) as a cheap signature-evasion rung.	A naive WG-signature block is evaded with LWO; tunnel passes traffic. `[evidence]`	`v02-data-plane/transport-lwo.md`	MVP
HVPN-FR-006	The Client MUST automatically escalate transports on handshake failure: plain → LWO →	DoD#4: when plain WG is blocked, the client auto-escalates to MASQUE without user action. `[evidence]`	`v02-data-plane/transport-selection-ladder.md`	MVP

ID	Statement	Acceptance criterion	Owning doc	Priority
	MASQUE/QUIC (→ Shadowsocks → UoT in P2).			
HVPN-FR-007	The Client MUST allow the user to manually pin a transport (overriding the auto-ladder).	`helix_pin_transport(kind)` pins; `None` restores auto.	`03` (FFI), `v04-client/helix-core-rust.md`	MVP
HVPN-FR-008	The edge MUST support port-based evasion (custom WG port / :443 / :53) and port-hopping.	Tunnel establishes on :443/udp and :53; a multi-listener accepts. `[evidence]`	`v02-data-plane/transport-masque-quic.md`, `01` (edge / data-plane)	MVP
HVPN-FR-009	The Client MUST support Shadowsocks-wrapped WG (AEAD) as an obfuscation rung.	WG-in-Shadowsocks tunnel passes traffic under a QUIC/UDP-hostile network. `[evidence]`	`v02-data-plane/transport-shadowsocks.md`	P2
HVPN-FR-010	The Client MUST support UDP-over-TCP as the last-resort transport when UDP is fully blocked.	Tunnel survives a total UDP block via UoT. `[evidence]`	`v02-data-plane/transport-udp-over-tcp.md`	P2
HVPN-FR-011	The auto-ladder SHOULD use per-network memory + regional priors to pick the likely-working transport first.	The ladder records last-working transport per network and tries it first.	`v02-data-plane/transport-selection-ladder.md`	P2
HVPN-FR-012	The edge MUST support MASQUE CONNECT-IP (RFC 9484) as a native IP-over-HTTP/3 datapath (advanced, no inner WG).	CONNECT-IP path passes IP packets; interop verified. `[evidence]`	`01` (CONNECT-IP), `transport-masque-quic.md`	P2
HVPN-FR-013	The Client MUST be able to use the Gateway as a plain privacy exit (full-tunnel to the internet), the Mullvad use case.	Full-tunnel routes all traffic via the Gateway exit; egress IP is the Gateway's. `[evidence]`	`01`, `v04-client/helix-core-rust.md`	MVP
HVPN-FR-014	The Client MUST support split tunneling (per-route, and per-app on Android/desktop).	A bypassed route/app egresses outside the tunnel; tunneled routes stay in. `[evidence]`	`03`, `v04-client/helix-core-rust.md`	MVP
				MVP

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-015	The Client MUST recover the tunnel automatically after a transient carrier drop, preserving the selected transport.	A simulated drop reconnects without user action; transport preserved. `[evidence]`	`v02-data-plane/orchestrator-and-state.md`	
HVPN-FR-016	The data plane MUST manage MTU correctly per transport (e.g. 1420 for plain WG; reduced under MASQUE).	No fragmentation/black-hole; large-payload transfer succeeds on each transport. `[evidence]`	`v02-data-plane/transport-plain-udp.md`, `wireguard-core.md`	MVP
HVPN-FR-017	The data plane SHOULD support DAITA (constant packet sizing + cover traffic) as an orthogonal shaping layer **above** WG.	DAITA shaping produces constant-size packets; measured overhead within budget. `[evidence]`	`v02-data-plane/daita.md`	P2
HVPN-FR-018	The Client MUST present connection state transitions (Disconnected/ Connecting(transport)/ Connected/ Reconnecting/Failed) to the UI via a hot status stream.	`helix_status_stream` emits each transition; UI is a pure function of it. `[evidence]`	`03` (§7.3 FFI), `v04-client/state-management.md`	MVP
HVPN-FR-019	The transport topology MUST ship a **fixed asymmetric per-leg default in MVP** — user?gateway runs the obfuscation ladder, gateway?connector is plain WG/UDP; **map-driven configurable per-leg transport policy is P2.**	MVP: the default asymmetric legs are honoured (user?gw escalates the ladder, gw?connector is plain WG) with no per-leg config surface. P2: per-leg transport policy is overridable from the pushed network map. `[evidence]`	`01` (D6), `v03-control-plane/svc-coordinator.md`	MVP (fixed default) / P2 (configurable
HVPN-FR-020	The data plane MUST support direct P2P + NAT traversal with a DERP-style relay fallback.	A P2P session forms across a real NAT; relay fallback engages when hole-punching fails. `[evidence]`	`01`, `v02-data-plane/routing-and-addressing.md`	P2

B. Identity & enrollment (FR-1xx)

Owning docs: `../04-security-privacy-pki.md`, `../v05-security/identity-and-enrollment.md`, `../v03-control-plane/svc-identity.md`, `../v03-control-plane/svc-pki.md`. Parity rows F15, F17.

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-101	The control plane MUST support OIDC identity for human principals (RBAC roles `admin` / `operator` / `member`, per `svc-identity.md`).	OIDC login yields a tenant-scoped identity with the correct RBAC role. `[evidence]`	`svc-identity.md`	MVP
HVPN-FR-102	The control plane MUST support anonymous device-token enrollment (no email / no PII) alongside OIDC.	A device enrolls and connects with no PII captured. `[evidence]`	`svc-identity.md`, `identity-and-enrollment.md`	MVP
HVPN-FR-103	A device MUST generate its WireGuard keypair on-device; the private key MUST NEVER leave the device.	The `pki`/registry schema has no private-key column; only the 32-byte public key is transmitted. `[evidence]` schema + wire capture.	`identity-and-enrollment.md`, `svc-pki.md`	MVP
HVPN-FR-104	Enrollment MUST be gated by a device cert + enrollment token.	An unenrolled/ untokened device is refused; a valid token enrolls. `[evidence]`	`svc-identity.md`, `identity-and-enrollment.md`	MVP
HVPN-FR-105	Every agent MUST authenticate the control channel with a short-lived (≤24 h, auto-renew) tenant-CA-signed mTLS device cert.	A control RPC with an expired/ absent cert is rejected; auto-renew succeeds before expiry. `[evidence]`	`svc-pki.md`, `pki-and-certs.md`	MVP
HVPN-FR-106	The control channel and data channel MUST share no key	Compromising one channel does not authenticate	`04` (§1.2), `svc-pki.md`	MVP

ID	Statement	Acceptance criterion	Owning doc	Priority
	material (cert compromise ≠ WG key; WG-key theft ≠ control auth).	the other (tested). `[evidence]`		
HVPN-FR-107	An admin MUST be able to see and revoke a device; revocation MUST be enforced at the edge within p99 < 1 s.	DoD#6: device revoke enforced <1 s — WG peer dropped, cert revoked, edge verdict removed atomically. `[evidence]`	`svc-registry.md`, `svc-pki.md`, edge	MVP
HVPN-FR-108	The CA hierarchy MUST be per-tenant; the tenant CA key is the single root secret to protect.	Each tenant has its own CA; cross-tenant cert validation fails. `[evidence]`	`svc-pki.md`, `pki-and-certs.md`	MVP
HVPN-FR-109	Device certs MUST rotate automatically without user interaction or tunnel interruption.	A cert rotation occurs mid-session with no tunnel drop. `[evidence]`	`pki-and-certs.md`	MVP
HVPN-FR-110	The system MUST support multi-tenant identity isolation enforced at the database (RLS), not only the app layer.	A cross-tenant read is denied by Postgres RLS even with app-layer bypass attempt. `[evidence]`	`data-model-ddl.md`, `svc-identity.md`	MVP
HVPN-FR-111	Enrollment MUST be available to all three system roles (Client, Connector, and — for HA — additional edges).	Both a Connector and a Client enroll (DoD#2). `[evidence]`	`svc-registry.md`	MVP

C. Policy & authorization (FR-2xx)

Owning docs: `../v03-control-plane/svc-policy.md`, `../v05-security/zero-trust-and-default-deny.md`. Principle P-zero-trust; parity row (no-logging adjacency).

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-201	Policy MUST be default-deny and fail-closed: the empty policy denies all; a compiler error denies rather than opens.	An unpoliced device has empty `AllowedIPs` and no verdict entry → reaches nothing. `[evidence]`	`svc-policy.md`, `zero-trust-and-default-deny.md`	MVP
HVPN-FR-202	The policy compiler MUST accept a Tailscale-ACL-flavored model (`group:X → net:Y:port/proto`).	A sample ACL compiles to per-peer `AllowedIPs` + verdict map. `[evidence]`	`svc-policy.md`	MVP
HVPN-FR-203	Policy MUST compile to two enforcement artefacts per device: WG `AllowedIPs` and an edge verdict map keyed `(src_overlay_ip, dst_cidr, l4proto, dport)`.	Both artefacts emitted; an allowed flow passes, a denied flow drops at the edge. `[evidence]`	`svc-policy.md`, `01` (§8 verdict map)	MVP
HVPN-FR-204	A Client MUST reach an authorized LAN host AND be denied an unauthorized one.	DoD#3: authorized host reachable, unauthorized denied (default-deny proven). `[evidence]`	`svc-policy.md`, edge	MVP
HVPN-FR-205	A policy edit MUST take effect within p99 < 1 s with no service restart.	DoD#5: an ACL change is enforced <1 s, no restart. `[evidence]`	`svc-policy.md`, `svc-coordinator.md`	MVP
HVPN-FR-206	Split horizon MUST be the default: Connectors cannot reach each other and Clients cannot reach un-granted networks unless policy says so.	Two connectors are mutually unreachable absent a rule; a client is denied a non-granted network. `[evidence]`	`svc-policy.md`, `routing-and-addressing.md`	MVP
HVPN-FR-207	Map distribution MUST be need-to-know: a device's network map contains only the peers/routes its policy already grants	A device's `WatchNetworkMap` snapshot omits non-granted peers. `[evidence]`	`svc-coordinator.md`, `04` (S3)	MVP

ID	Statement	Acceptance criterion	Owning doc	Priority
	(filtered server-side, before the wire).			
HVPN-FR-208	Policy SHOULD be expressible as policy-as-code / GitOps.	A policy committed to a repo reconciles into the control plane. `[evidence]`	`svc-policy.md`	P2

D. Routing, addressing & multi-network (FR-3xx)

Owning docs: [`../v02-data-plane/routing-and-addressing.md`](../v02-data-plane/routing-and-addressing.md), [`../v03-control-plane/svc-ipam.md`](../v03-control-plane/svc-ipam.md). Differentiators X1/X2 [00 §9], multi-network model [00 §4.2], D4.

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-301	The Gateway MUST give one user policy-scoped access to N joined private networks (the 1→N overlay).	A single user reaches granted hosts across ≥2 distinct connector networks. `[evidence]`	`routing-and-addressing.md`, `svc-policy.md`	MVP
HVPN-FR-302	The system MUST resolve overlapping RFC1918 ranges across connectors so two networks both exposing `192.168.1.0/24` never collide.	Two connectors advertising the same CIDR are reachable as distinct address spaces. `[evidence]`	`routing-and-addressing.md`, `svc-ipam.md`	MVP
HVPN-FR-303	The overlay MUST assign each tenant an IPv6 ULA /48 and map advertised IPv4 LANs into it via 4via6 (D4 recommendation), with per-network NAT as a documented fallback.	An advertised IPv4 LAN is reachable through its 4via6 mapping; NAT fallback documented + testable. `[evidence]`	`svc-ipam.md`, `routing-and-addressing.md`	MVP
			`svc-ipam.md`	MVP

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-304	Each node MUST receive a stable overlay IP.	A node's overlay IP persists across reconnects. `[evidence]`		
HVPN-FR-305	A Connector MUST advertise the CIDRs its network exposes (`route.advertised`), and the Gateway MUST route authorized client traffic into that LAN and responses back out.	DoD#3 path: client → gateway → connector LAN host and back. `[evidence]`	`svc-registry.md`, edge, `routing-and-addressing.md`	MVP
HVPN-FR-306	The Gateway MUST route between the network-side leg (Connector→Gateway) and the user-side leg (Client→Gateway) — the "two-way" model — without either side opening an inbound port.	Both legs are outbound; the Gateway stitches them; no inbound listener on any private network. `[evidence]`	`01` (edge / data-plane), `v04-client/helix-core-rust.md` (advertise/route mode), `svc-registry.md`	MVP
HVPN-FR-307	The IPAM service MUST allocate from the tenant overlay pool and track host allocations + 4via6 mappings.	Pool allocation is collision-free under concurrent enrollment. `[evidence]` stress test.	`svc-ipam.md`	MVP

E. Multi-hop (FR-4xx)

Owning docs: `../v02-data-plane/multihop.md`, `02`/`03`. Parity row F10.

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-401	The Client MUST support multi-hop (entry/exit separation) via nested WireGuard with per-hop keys.	A two-hop path forms; entry and exit use separate keys; egress IP is the exit's. `[evidence]`	`multihop.md`	P2
				P2

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-402	Multi-hop path selection MUST be control-plane-orchestrated (entry/exit chosen via the network map).	The coordinator delivers the entry/exit pair; the client establishes the nested tunnels. `[evidence]`	`multihop.md`, `svc-coordinator.md`	
HVPN-FR-403	Multi-hop SHOULD allow entry/exit jurisdiction separation (entry and exit in different regions).	Entry and exit resolve to different regions when configured. `[evidence]`	`multihop.md`	P2

F. Kill-switch & leak protection (FR-5xx)

Owning docs: `../v05-security/kill-switch-and-dns-leak.md`, `03`. Parity rows F11/F13; invariant S9.

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-501	The kill-switch MUST be core-owned state (driven by `helix-core`'s state machine), not hand-edited firewall rules.	The kill-switch is set/cleared by the core state machine; no manual rule editing path. `[evidence]`	`kill-switch-and-dns-leak.md`	MVP
HVPN-FR-502	No plaintext egress MUST occur when the tunnel is down OR while escalating transports.	DoD#7: on tunnel drop, no packet leaks outside the tunnel. `[evidence]` leak capture.	`kill-switch-and-dns-leak.md`	MVP
HVPN-FR-503	The Client MUST force DNS through the tunnel and block plaintext :53 off-tunnel (DNS-leak protection).	DoD#7: DNS queries resolve only through the tunnel; off-tunnel :53 is blocked. `[evidence]`	`kill-switch-and-dns-leak.md`	MVP
HVPN-FR-504	The kill-switch MUST be implemented per-OS via the OS firewall, driven by the same core state machine across platforms.	Each supported OS enforces the kill-switch via its firewall from the shared state machine. `[evidence]`	`kill-switch-and-dns-leak.md`, platform shims	MVP

G. Console & administration (FR-6xx)

Owning docs: `../v03-control-plane/svc-api.md`, `../v04-client/web-console.md`, `../v05-security/audit-and-compliance.md`. Persona: business-admin / tenant-owner [00 §5.3].

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-601	The Console MUST provide CRUD for tenants, users, devices, networks, routes, and policies.	Each entity is creatable/ readable/ updatable/ deletable subject to RBAC. `[evidence]`	`svc-api.md`, `web-console.md`	MVP
HVPN-FR-602	The Console MUST be an API-client-only build with no tunnel core (`runHelixApp(Console, …)` omits `core_ffi`).	The Console build links no Rust tunnel core; runs in browser + desktop from the same Flutter tree. `[evidence]`	`web-console.md`, `03`	MVP
HVPN-FR-603	The Console MUST receive live state updates (WS/SSE) — no polling for topology/state.	A policy/ device change pushes to the Console live. `[evidence]`	`svc-api.md`	MVP
HVPN-FR-604	The Console MUST present a live topology view of tenants/networks/ devices/routes.	The topology view reflects current state and updates on change. `[evidence]`	`web-console.md`	MVP
HVPN-FR-605	The Console MUST surface an audit trail of **control** actions (who-did-what to identity/policy/ devices), never destinations/flows.	`audit_events` shows control mutations; no traffic destinations recorded. `[evidence]` schema + UI.	`audit-and-compliance.md`, `svc-telemetry.md`	MVP
HVPN-FR-606	The Console MUST support multi-tenant management with strict per-tenant isolation.	An admin of tenant A cannot see tenant B's resources. `[evidence]`	`web-console.md`, RLS	MVP

P3

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-607	The Console MAY provide optional multi-tenant billing.	Billing flows function for the managed SKU when enabled; absent by default.	`web-console.md`	
HVPN-FR-608	The Console MUST be responsive across phone/tablet/desktop and ship light + dark themes from the OpenDesign system.	Visual-regression suite passes light+dark; no element overlap/overlay (§11.4.162). `[evidence]`	`web-console.md`, Volume 10	MVP

H. Connector (FR-7xx)

Owning docs: `../v04-client/helix-core-rust.md` (advertise/route mode) + Volume 4 platform shims (`shim-*.md`) + `../v03-control-plane/svc-registry.md` (there is no standalone "Connector spec" file — the Connector is the same `helix-core` in advertise/route mode plus its registry contract). Differentiator X2; persona connector-operator [00 §5.2].

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-701	The Connector MUST dial outbound to the Gateway and MUST NOT require any inbound port-forward.	The Connector establishes the tunnel with zero inbound exposure on its network. `[evidence]`	`helix-core-rust.md` (advertise/route mode), `svc-registry.md`	MVP
HVPN-FR-702	The Connector MUST run headless (daemon) with an optional slim config UI.	The daemon runs without a UI; the optional UI configures it. `[evidence]`	`helix-core-rust.md` (advertise/route mode)	MVP
HVPN-FR-703	The Connector MUST share the same Rust `helix-core` as the Client, in advertise/route	The Connector links the same crate; runs in advertise/route mode.	`helix-core-rust.md`	MVP

ID	Statement	Acceptance criterion	Owning doc	Priority
	mode (not capture mode).			
HVPN-FR-704	The Connector MUST advertise its network's CIDRs to the Gateway and route authorized traffic into the LAN.	Advertised CIDR appears in the registry; authorized client reaches a LAN host. `[evidence]`	`svc-registry.md`, `01` (edge)	MVP
HVPN-FR-705	The Connector SHOULD support local ACLs scoped to its own network.	A local ACL on the connector is honoured for its network. `[evidence]` (`UNVERIFIED` interaction with central policy — fixed by `svc-policy.md`)	`svc-policy.md`, `helix-core-rust.md` (advertise/route mode)	MVP
HVPN-FR-706	The Connector MUST be runnable on Android/ embedded appliance hardware in addition to Linux/Windows/ macOS.	A Connector build runs on an embedded/Android target. `[evidence]`	`helix-core-rust.md` (advertise/route mode), `shim-android.md`	P2
HVPN-FR-707	The Connector MUST follow availability-following on drop: detect, log offline, reconnect with defined timings, resume (§11.4.144 alignment).	A simulated drop is logged offline, reconnected, and resumed with no silent gap. `[evidence]`	`orchestrator-and-state.md`	MVP

I. Observability & telemetry (FR-8xx)

Owning docs: `../v03-control-plane/svc-telemetry.md`, `../v06-deploy/observability.md`, `../v05-security/no-logging-as-code.md`. Principle P7; parity row F14.

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-801	The data plane MUST persist only aggregate counters + ephemeral routing/presence state — NO durable connection/traffic/packet table.	DoD#8: a CI schema-lint fails the build if a connection-log-shaped durable table appears; runtime check confirms none. `[evidence]`	`no-logging-as-code.md`, `data-model-ddl.md`	MVP
HVPN-FR-802	Live presence/routing state MUST live in TTL'd Redis (ephemeral), never a durable table.	Presence entries expire on TTL; no durable persistence of presence. `[evidence]`	`svc-events.md`, `no-logging-as-code.md`	MVP
HVPN-FR-803	The control plane MUST expose health + counters as Prometheus metrics.	Prometheus scrapes counters/health; metrics carry no flow/destination data. `[evidence]`	`svc-telemetry.md`, `observability.md`	MVP
HVPN-FR-804	The system MUST expose convergence + event-lag SLOs as observable metrics.	Convergence p99 and event-lag are measurable from metrics. `[evidence]`	`observability.md`, NFR doc	MVP
HVPN-FR-805	Telemetry MUST be derivable as counts only (no flows) for product success metrics.	Success-metric dashboards use counts; no per-flow data exists to query. `[evidence]`	`svc-telemetry.md`, `success-metrics.md`	MVP

J. Deployment & self-host (FR-9xx)

Owning docs: `../05-repo-layout-tooling-and-helix-ecosystem.md`, `../v06-deploy/helixvpnctl.md`, `../v06-deploy/podman-quadlets.md`, `../v06-deploy/ha-and-multiregion.md`. Principle P6; differentiator X3.

ID	Statement	Acceptance criterion	Owning doc	Priority
				MVP

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-901	A self-hoster MUST stand up a Gateway from zero on a clean VPS with one `helixvpnctl init`.	DoD#1: `helixvpnctl init` brings up edge + control + Postgres + Redis in one rootless Podman pod. `[evidence]`	`helixvpnctl.md`, `podman-quadlets.md`	
HVPN-FR-902	Deployment MUST be rootless Podman (one pod for homelab), per §11.4.161.	The pod runs rootless; no root/ sudo escalation. `[evidence]`	`podman-quadlets.md`	MVP
HVPN-FR-903	The same OCI image set MUST scale to an HA multi-region fleet without a code rewrite (deployment topology change only).	The same images deploy as N stateless coordinators + Patroni PG + NATS. `[evidence]`	`ha-and-multiregion.md`	P2
HVPN-FR-904	`helixvpnctl` MUST provide init / key management / enrollment-token / policy / revoke commands.	Each subcommand functions end-to-end. `[evidence]`	`helixvpnctl.md`	MVP
HVPN-FR-905	Deployment MUST also be expressible as Docker Compose and Kubernetes manifests.	Equivalent Compose + K8s manifests bring up the stack. `[evidence]`	`docker-compose.md`, `kubernetes.md`	P2
HVPN-FR-906	The edge/data-plane container MUST be hardened: read-only rootfs, seccomp, `NET_ADMIN`-only, no SSH (S8).	The container runs with only `NET_ADMIN`, read-only rootfs, seccomp profile; no SSH. `[evidence]`	`podman-quadlets.md`, `04` (S8)	MVP
HVPN-FR-907	The system MUST be fail-static: if the control plane is down, existing tunnels keep forwarding.	DoD-adjacent: control plane stopped mid-session → existing tunnel keeps	`01` (edge / data-plane), `02`	MVP

ID	Statement	Acceptance criterion	Owning doc	Priority
		forwarding. `[evidence]`		
HVPN-FR-908	Any gitignored build-essential artefact MUST have a documented regeneration/re-obtain mechanism (§11.4.77).	A fresh clone regenerates required artefacts via `scripts/setup.sh`. `[evidence]`	`repo-layout-and-decoupling.md`	MVP

K. Clients & platform apps (FR-10xx)

Owning docs: `../03-client-core-and-ui.md` and Volume 4 (`helix-core-rust.md`, `ffi-surface.md`, `helix-ui-flutter.md`, `shim-*.md`). Differentiator X5; persona privacy-consumer/end-user.

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-1001	The three apps (Access, Connector, Console) MUST build from one Flutter tree via `runHelixApp(flavor, home, capabilities)`.	All three flavors build from one tree; capability gating differs per flavor. `[evidence]`	`helix-ui-flutter.md`	MVP
HVPN-FR-1002	Access + Connector MUST share one Rust `helix-core`; UI is a pure function of the FFI status stream.	Both apps link the same core; UI state derives from `helix_status_stream`. `[evidence]`	`helix-core-rust.md`, `ffi-surface.md`, `state-management.md`	MVP
HVPN-FR-1003	The Access app MUST ship a one-button connect experience (Mullvad-style).	One tap connects with auto-obfuscation; minimal user decisions. `[evidence]`	`helix-ui-flutter.md`, Volume 10	MVP
HVPN-FR-1004	The iOS `NEPacketTunnelProvider` MUST run the Rust core under the platform memory ceiling with ≥30% headroom.	G3 (make-or-break): on-device RSS profile shows ≥30% headroom under the ~15 MB ceiling. `[evidence]`	`shim-apple.md`, `03`	MVP
HVPN-FR-1005	The Android app MUST integrate `VpnService` + JNI (builder/protect/fd handoff) with background-kill resilience.	The app tunnels on Android; survives background; fd handoff correct. `[evidence]`	`shim-android.md`	MVP
HVPN-FR-1006	The Linux client MUST integrate kernel WG/tun + systemd.	The Linux client tunnels via kernel WG. `[evidence]`	`shim-linux.md`	MVP
HVPN-FR-1007	The Windows client MUST integrate `wireguard-nt/`	The Windows app tunnels via wireguard-	`shim-windows.md`	P2

ID	Statement	Acceptance criterion	Owning doc	Priority
	wintun + a privileged service with named-pipe IPC and WFP split-tunnel.	nt; split-tunnel via WFP works. `[evidence]`		
HVPN-FR-1008	The macOS client MUST integrate the Network Extension (`NEPacketTunnelProvider`) desktop app.	The macOS app tunnels via NE. `[evidence]`	`shim-apple.md`	P2
HVPN-FR-1009	The Access app MUST build and tunnel on HarmonyOS NEXT (OpenHarmony Flutter fork, Network Kit VPN ability).	On-device: the app tunnels on HarmonyOS NEXT. `[evidence]` (biggest platform risk)	`shim-harmonyos.md`	P3
HVPN-FR-1010	The Access app MUST build and tunnel on Aurora OS (omprussia Flutter, Qt/C++ tun shim, signed RPM).	On-device: the app tunnels on Aurora. `[evidence]`	`shim-aurora.md`	P3
HVPN-FR-1011	The Web build MUST be Console (management) only, with an OPTIONAL in-page WASM MASQUE proxy that proxies the **browser's own** traffic — never a system VPN.	The Web build manages the system; the WASM proxy (when present) reaches a joined network from the browser only. `[evidence]`	`web-console.md`	P3
HVPN-FR-1012	The Access app MUST let the user pick exit/network and toggle obfuscation.	Exit/network selection + obfuscation toggle function. `[evidence]`	`helix-ui-flutter.md`, `ffi-surface.md`	MVP
HVPN-FR-1013	All client apps MUST ship light + dark theme variants per the OpenDesign system, with no element overlap/ overlay.	Visual-regression light+dark passes (§11.4.162). `[evidence]`	Volume 10	MVP
HVPN-FR-1014	All three apps MUST drive the system end-to-end (DoD#8 second clause).	DoD#8: Access + Connector + Console all drive the live system. `[evidence]`	all client docs	MVP

L. Post-quantum & advanced privacy (FR-11xx)

Owning docs: `../v05-security/post-quantum.md`, `01`/`02`. Parity row F16; invariant S10.

ID	Statement	Acceptance criterion	Owning doc	Priority
HVPN-FR-1101	The handshake MUST support a	A PQ-enabled handshake	`post-quantum.md`	P2

ID	Statement	Acceptance criterion	Owning doc	Priority
	post-quantum KEM (ML-KEM / FIPS-203) deriving a PSK mixed into the classical WG handshake.	interops and is measured; the PSK is mixed in. `[evidence]`		
HVPN-FR-1102	PQ MUST be hybrid, never PQ-only: with PQ off, classical WG remains secure; with PQ on, an attacker must break both.	Disabling PQ leaves a secure classical tunnel; enabling adds the PQ layer. `[evidence]`	`post-quantum.md`	P2
HVPN-FR-1103	The system MAY evaluate Rosenpass as an alternative PQ mechanism.	A documented evaluation of Rosenpass vs ML-KEM PSK exists.	`post-quantum.md`	P2

M. MVP Definition-of-Done traceability

The 8-criteria MVP DoD [00 §10.2], [SPEC §8.1] maps onto the FRs above. Every DoD criterion MUST be satisfied with captured runtime evidence (§11.4.69); none may be a metadata-only PASS.

DoD #	Criterion	Satisfying FR(s)
DoD-1	Self-host from zero via `helixvpnctl init`	FR-901, FR-902
DoD-2	Enroll a Connector AND a Client	FR-104, FR-111, FR-701
DoD-3	Reach an authorized LAN host AND deny an unauthorized one	FR-204, FR-201, FR-305, FR-306
DoD-4	Auto-escalate to MASQUE when plain WG is blocked	FR-006, FR-004
DoD-5	Policy edit reflected < 1 s, no restart	FR-205
DoD-6	Device revoke enforced < 1 s	FR-107
DoD-7	Kill-switch + DNS-leak protection (no leak on drop)	FR-501, FR-502, FR-503, FR-504
DoD-8	No durable connection log AND all three apps drive the system	FR-801, FR-1014

N. Parity-matrix → FR coverage

Every Mullvad-parity row [00 §9] and HelixVPN-beyond-Mullvad row maps to ≥1 FR, so the parity matrix is the acceptance checklist and nothing is orphaned.

Parity row	Feature	FR(s)	Priority
F1	WireGuard-only crypto	FR-001	MVP
F2	QUIC obfuscation (MASQUE)	FR-004	MVP
F3	MASQUE CONNECT-IP	FR-012	P2
F4	Shadowsocks	FR-009	P2
F5	UDP-over-TCP	FR-010	P2
F6	LWO	FR-005	MVP
F7	Automatic obfuscation	FR-006	MVP
F8	Custom WG port / 443 / 53	FR-008	MVP
F9	DAITA	FR-017	P2
F10	Multi-hop	FR-401, FR-402, FR-403	P2
F11	Kill-switch	FR-501, FR-502, FR-504	MVP
F12	Split tunneling	FR-014	MVP
F13	DNS leak protection	FR-503	MVP
F14	No-logging	FR-801, FR-802	MVP
F15	Anonymous account (no email)	FR-102	MVP
F16	Post-quantum handshake	FR-1101, FR-1102	P2
F17	Per-device management	FR-107	MVP
X1	Multi-network overlay (1 user → N nets)	FR-301, FR-302, FR-303	MVP
X2	Outbound-only network onboarding	FR-701, FR-306	MVP
X3	Self-hostable Mullvad-class stack	FR-901, FR-903	MVP
X4	Direct P2P + NAT traversal + relay	FR-020	P2
X5	8-platform reach incl. Aurora + HarmonyOS	FR-1004..FR-1011	MVP/P2/P3

> **Coverage honesty (§11.4.118).** This FR set covers the overview's enumerated capability set (parity matrix F1–F17, differentiators X1–X5, the 8 DoD criteria, and the seven principles). It is *necessary, not provably exhaustive*: a capability introduced by a later Volume 2–6 doc that the overview did not enumerate becomes a new FR via the §11.4.93 workable-item path. FRs whose detailed contract is fixed by an owning doc that has not yet pinned it (e.g. FR-705 local-ACL scope) are marked `UNVERIFIED` inline. No requirement quotes a quantitative target not present in the overview/spine/security-overview.

Sources verified

- `00-product-scope-and-principles.md` §8 (seven principles → FR-001/201/801/901/907), §9 (parity matrix F1–F17, X1–X5 → §N), §10.2 (8-criteria MVP DoD → §M), §3/§4 (roles + two-way/multi-network → FR-3xx/7xx), §11 (decisions D1/D4/D6 → FR-004/303/019).
- `SPECIFICATION.md` §7 (the three cross-cutting contracts: `Transport` trait → FR-003, `WatchNetworkMap` → FR-205/207, FFI → FR-018), §8 (phase roadmap + gates G1–G6 → FR-002/004/1004, DoD → §M), §9 (decision register).
- `04-security-privacy-pki.md` §0.1 (S1–S11 → FR-103/106/107/201/207/801/906), §1.2/§1.3 (auth channels + default-deny).
- `MASTER_INDEX.md` Volume 1 row (declared scope: "Every FR, numbered (HVPN-FR-NNN), with acceptance"); owning-doc filenames cross-referenced from the Volume 2/3/4/5/6/10 rows.

Honesty note (§11.4.6): every quantitative acceptance target (≥80%/≥50% throughput, p99 < 1 s convergence/revocation, ~15 MB / ≥30% memory headroom) is quoted from the overview/spine/security-overview, never invented. FRs whose detailed contract awaits an owning doc are marked `UNVERIFIED`. `[evidence]` flags criteria requiring captured runtime evidence per §11.4.69 — none may pass on metadata/config-only signal.

Constitution bindings: §11.4.44 (revision header), §11.4.6 (no-guessing — UNVERIFIED on un-pinned contracts; targets quoted not invented), §11.4.69 (captured-evidence acceptance), §11.4.93 (each FR → workable item), §11.4.118 (coverage honesty), §11.4.169 (acceptance tied to mandatory test types, Volume 8), §11.4.65/.153 (HTML+PDF[+DOCX] exports follow in refinement).